

VISUAL REASONING	LOGICAL DERIVATION	NUMERICAL ABILITY
	GENERAL KNOWLEDGE, ARCHITECTURE & DESIGN	
DESIGN THINKING	LANGUAGE INTERPRETATION	DESIGN SENSITIVITY

Identify the image with the shadow cast by the buildings at 09.00 AM and 03.00 PM.

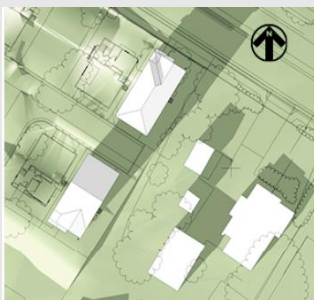
A.



C.



B.



D.



Source: <http://www.whablog.com/>

A.	09.00 AM – Options 1,4	C.-	09.00 AM- Options 2,3
	03.00 PM – Options 2,3		03.00 PM – Options 1,4
B.	09.00 AM – Options 1,2	D.	09.00 AM- Options 1,3
	03.00 PM – Options 3,4		03.00 PM – Options 2,4

Answer key:

Theory:

Shadows are the relatively dark figures cast upon a surface by an opaque body or part of a body intercepting the rays from a light source.

The shape of the shadow is dependent on

- The position of shade line & the observer, The direction of the light and The form of the surfaces on which the plane of the shadow falls
- When the object is not in contact with the ground, the shadow will be seen detached from the object

A.	09.00 AM – Options 1,4
	03.00 PM – Options 2,3

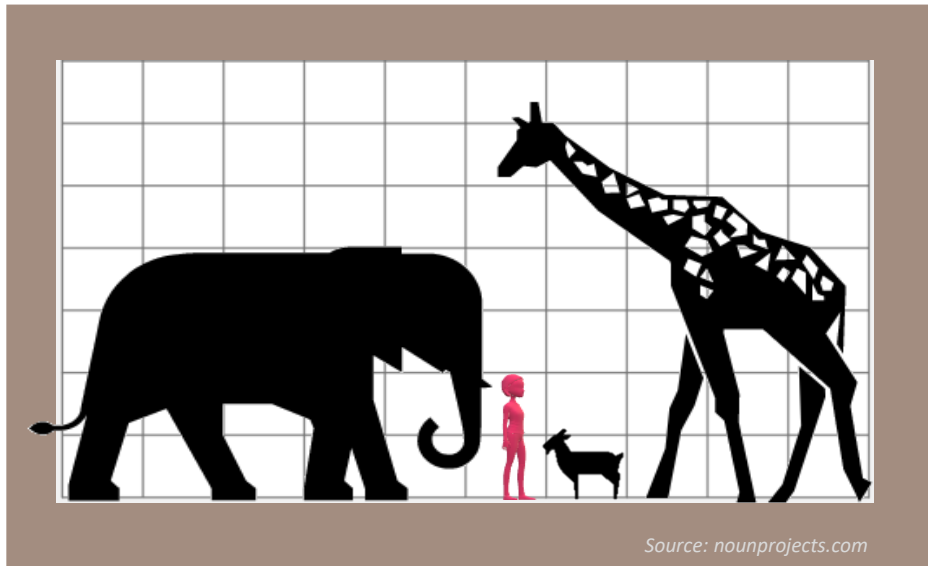
Suggested Reading: Basics of light and shadow

" For a swift review of the subject, kindly consult the study materials available on the website"

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Mona went to zoo as a part of school excursion. She was so excited to see all animals and was describing them to her mom back home. She narrated the animal's height by comparing them to her height. Based on the graphics, find the correct proportions as recited by her.



A	▪ Elephant -2.5 times of Mona ▪ Goat-Same height of Mona ▪ Giraffe-4 times of Mona	C	▪ Elephant-3 times of Mona ▪ Goat-1/2 height of Mona ▪ Giraffe-2 times of Mona
B	▪ Elephant -2 times of Mona ▪ Goat - 1/2 height of Mona ▪ Giraffe - 3 times of Mona	D	▪ Elephant-4 times of Mona ▪ Goat-1/2 height of Mona ▪ Giraffe - 4 times of Mona

Answer key:

Option B

- **Elephant – 2 times of Mona**
- **Goat –1/2 height of Mona**
- **Giraffe – 3 times of Mona**

Theory:

Proportion

Proportion refers to the relationship of one part to another or to the whole, or between one object and another. This relationship may be one of magnitude, quantity, or degree.

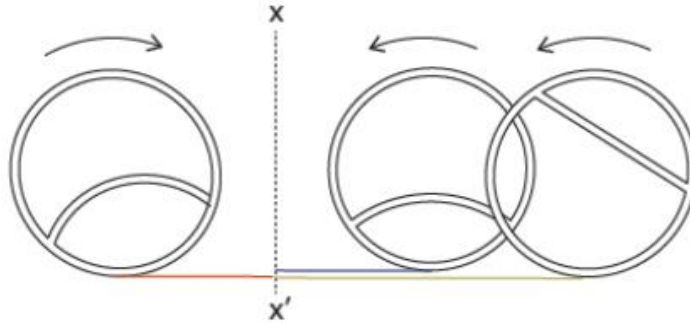
Suggested Reading: visual composition

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Three rings are shown in the figure. They roll in the specified directions and stop when centres of all the rings coincide at line XX' . What pattern would they make?



Answer key:



Theory:

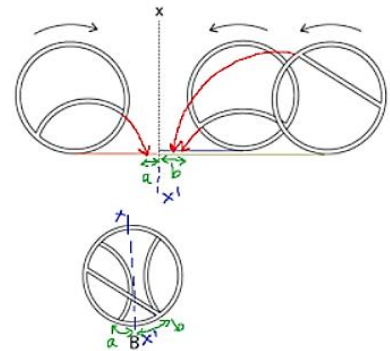


Figure shows the position of the points on the circles which will roll and position as shown by red arrows. In option B, if you draw a vertical section line $X-X'$, the position of the above points should be oriented at the same circular distance on either side, as shown in the bottom image.

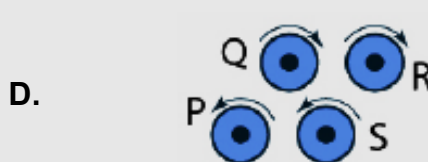
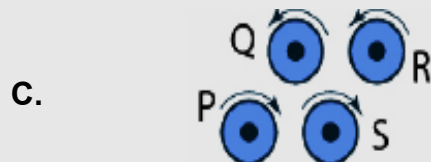
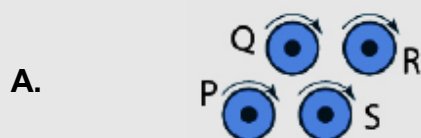
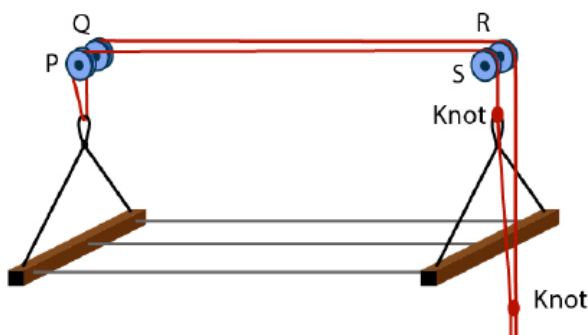
Suggested Reading: Geometry, Rotation

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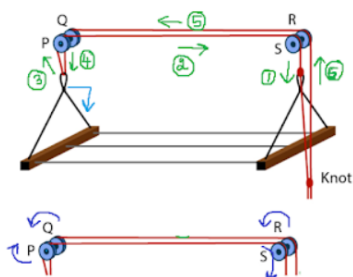
Figure shows a clothes line attached to four independent pulleys P, Q, R & S. Which of the options represent the correct rotation of pulleys that would result in lowering the clothes line?



Answer key:



Theory:



In the top image shown, blue arrows show the requirement of lowering the clothes line. The corresponding travel of the rope at different parts are shown by green arrows. Start at the knot with shown marked (circled) as 1, then move in the opposite direction of the rope travel as marked by numbered circles. Given the direction of the rope travel, it is now easy to identify the direction of rotation of the pulleys. This is seen in the second image. Refer both the images to mark the blue arrows.

Suggested Reading: Rotation and Direction

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The term _____ is defined as the construction of individual building units bonded together with mortar.



Source: istockphoto.com

A. Masonry

B. Coping

C. Brick

D. Roof

Answer key:

A. Masonry

Theory:

The term Masonry is used to indicate the art of building the structures in either stone, bricks or concrete blocks. Masonry is normally used for the construction of foundations, walls, columns and other similar structural components of buildings. Masonry is generally a highly durable form of construction.

Suggested Reading: Basics of building construction/Components

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Identify the Art movement related to the painting?



Source: <https://www.milanartinstitute.com/blog/4-tips-for-painting-like-an-impressionist>

A. Cubism

C. Dadaism

B. Impressionism

D. Renaissance

Answer key:

Theory:

The Starry Night, a moderately abstract landscape painting (1889) of an expressive night sky over a small hillside village, one of Dutch artist Vincent van Gogh's most celebrated works. Vincent van Gogh was a Expressionist painter whose work — notable for its beauty, emotion and color — highly influenced 20th-century art. People, places and objects are distorted or exaggerated. Even nature is sometimes distorted. The scenes show a modern world which is hostile and alienating. The sinister feeling is amplified by aggressive and raw brush strokes.

B. Impressionism

Suggested Reading: Art Movement

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What role does storytelling play in design thinking?

A. It restricts designers to a linear problem-solving approach.

B. It helps communicate design concepts and ideas effectively.

C. It encourages designers to disregard user feedback .

D. It limits creativity and innovation in the design process

Answer key:

B. It helps communicate design concepts and ideas effectively

Theory:

Storytelling in design thinking involves communicating design concepts, ideas, and solutions effectively through narratives. It helps engage stakeholders, convey the value of design solutions, and create a shared understanding of the problem and its potential solutions.

Suggested Reading: Design Thinking

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Read each sentence to find out whether there is any grammatical error in it. The error, if any will be in one part of the sentence. The letter of that part is the answer. If there is no error, the answer is 'D'. (Ignore the errors of punctuation, if any).

1. After finding the problem,
2. the mechanic cleaned the carburetor
3. started the engine
4. and found it working smoothly.

A. 1

C. 3

B. 2

D. 4

Answer key:

Theory:

A. 1

The correct option is A. 1

The sentence begins with 'after finding' which is the mistake in it. A sentence usually never begins with 'after finding' followed by the consequence of finding it. Whenever a sentence talks about the consequence of finding something, it must always begin with 'on finding'. Therefore the incorrect part of the sentence is A. 1. The other three parts 2, 3 and 4 are correct.

Suggested Reading: Grammar

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Read the passage and answer the question:

Have you ever heard someone sing, “Happy birthday to you . . . happy birthday to you . . .”? If so, you are hardly alone—millions of people sing that song every year, and the tune is one of the most familiar of any song in North America.

Ironically, it is not known who wrote it! The tune was written by two sisters, Patty and Mildred Hill, who were school teachers in the late 19th century. But their song had different words; instead of “happy birthday to you,” the song ran “good morning to all.” The two sisters would sing this ditty to their students each morning as school began.

In 1924, the tune was published in a song book, which added the “happy birthday” lyrics as verse two. Today, the second verse of that song has become one of the best known tunes in the world—and nobody knows who wrote the words!

Why is it ironic that the author of the words to “Happy Birthday” is not known?

- A. because almost everyone knows the song
- B. because it shows how unjust life can be.
- C. because of modern copyright laws
- D. because it was published in 1924

Answer key:

A. because almost everyone knows the song.

Theory:

A. Something is ironic if the result is the opposite of what one would expect. It is ironic that nobody knows who wrote this song because almost everyone knows the song itself—we would expect that author to be famous, too

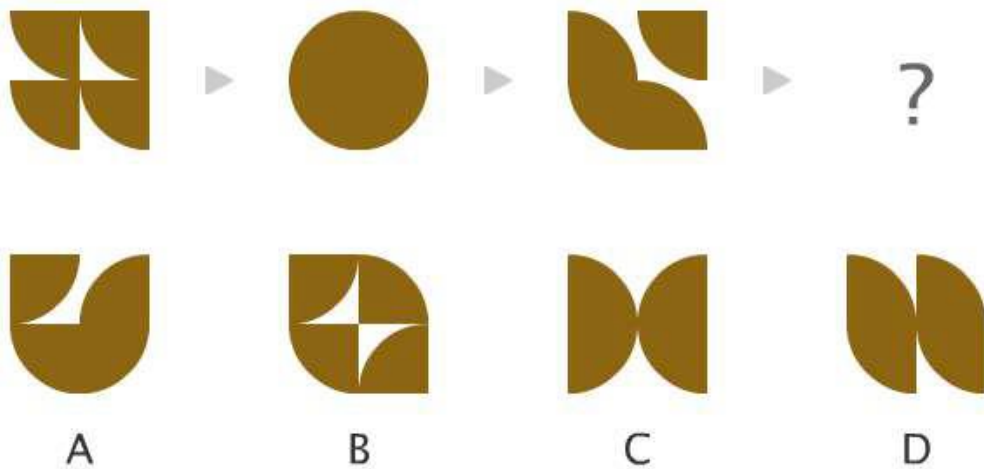
Suggested Reading: Passages and Interpretations

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Which option will replace the question mark?



A. A

C. C

B. B

D. D

Answer key:

Theory:

Option B:

The four quadrants are taken through a series of operations independently for it to form the consequent figure.

The bottom left quarter does not undergo any change.

The top left quadrant rotates clockwise 90 degrees each step

The top right rotates 180 degrees for every step.

The bottom right quadrant rotates 90 degrees anticlockwise.

These operations form the figure B.

Suggested Reading: Principles of design

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